



Why is mold fuzzy?

Have you ever wondered why mold appears to be fuzzy. Check out this writeup to find out. <https://dgit.in/sep19-03>



Global warming affects solar cells

The output of a solar cell decrease by approximately 0.45% with every single degree rise in temperature. <https://dgit.in/sep19-04>

as a human being. Many species of dwarf elephants were found all over the world, including in Crete, Sicily and Cyprus. The smallest of these was the pygmy elephant from Malta, which was only a meter in height. In all these islands, there were also species of dwarf hippopotami. The critically endangered tamaraw in the Philippines, is a water buffalo that stands just 40 inches tall. There are recorded cases of island dwarfism in homo sapiens as well. Perhaps the most well known of these is the *Homo floresiensis*, a species that has earned the moniker of “hobbits”. The species was only discovered in 2004. *Homo luzonensis*, another species yet to be confirmed, has only been discovered in 2007, and also exhibited island dwarfism. The native people of the Andaman islands are also considered an example of island dwarfism.

By and large, the changes in the sizes of carnivores are not as pronounced as the changes in the sizes of herbivores. Additionally, the extent to which island dwarfism occurs is closely tied to the existence, or lack of predators. In the absence of predators, dwarfism is more pronounced.

GIGANTISM

When birds colonise a new island, a well studied set of physical transformations tend to occur. Because of a lack of land animals in niches typically occupied by herbivores, the birds tend to grow into this gap. They lose the ability to fly, and gaining an advantage in the reduced metabolic costs. They also tend to grow larger. Well known examples of these include the moa and the dodo. Birds can also become the apex predators on islands. The Haast's eagle, with a wingspan of 2.6 meters and the Jamaican caracara are examples of such predators. In the ancient New Zealand ecosystem, think of the moa as playing the role of the grazing deer, while the Haast's eagle as the tiger at the top of the food chain.

While the general biogeographic trend is for smaller animals to grow larger, there are some exceptions. The

Kodiak bear, which lives in the Kodiak Archipelago in Alaska is one of these exceptions. It is a brown bear that is comparable in size to the polar bear, and is among the largest species of bears, as well as land carnivores. Island gigantism is thought to be related to diet and prey availability. Depending on the size and number of prey population, insular carnivores tend to grow larger or smaller. However, the diet of the Kodiak bear is similar to the diet of brown bears elsewhere, although it grows to large sizes.

There are two puzzling creatures when it comes to insular gigantism. These are that of the giant tortoises in the Galapagos Islands and the Komodo dragon. While these creatures are unusually large, scientists simply cannot figure out if they are cases of island gigantism or island dwarfism. This is because the fossil records show related species that were even larger! Unfortunately, most of the creatures known to exhibit island gigantism are either already extinct, or critically endangered.

TAMENESS

One of the advantages of gigantism is the number of predators that can kill larger creatures are reduced. Creatures living on islands without any predators, may unlearn the behaviors needed to avoid predators. Penguins for example, stay away from predators such as seals and killer whales in the seas. However, on land, they are curious about humans as they have no natural land predators.

Perhaps the best known example of island tameness is that of the dodo. The bird lived only in the remote Mascarene islands, and were ridiculously easy to catch, making them easy prey for sailors. The sailors would catch them for additional food on long sea voyages, while their eggs were consumed by ship animals such as pigs and dogs.

They were flightless, and unafraid of people. Habitat loss also contributed to the extinction of the bird. The dodo was extinct within two centuries of it first being sighted by western sailors.

One of the largest known birds that ever lived, the *Aepyornis maximus* was almost twice as tall as a human being, reaching a height of 3 meters. It could weigh up to 540 kg, and its eggs could reach 1 meter in diameter. The bird only lived in Madagascar, and its closest living relative is the Kiwi. Around 1000 AD, the bird is believed to have gone extinct, probably because of human activity.

Insular species are often unable to learn the required responses to predators quickly enough. Island tameness



The Stephens Island wren

Credit: NZ Birds Online

can decimate populations if new predator species are introduced. Opportunistic predators such as cats, and feral descendents of pets can wreak havoc in island ecosystems. One of the well-known cases is the extinction of Lyll's Wren, a small flightless bird in the remote Stephens Island in New Zealand. According to folklore, the bird was discovered and hunted to extinction by a single pet cat of the keeper of a lighthouse, named Tibbles. This is understood to be a myth, but feral cats were the cause of the extinction. The only conservation effort that can effectively preserve local populations is to strictly prevent the introduction of new species in an island environment. New Zealand is at the forefront of such conservation efforts.